

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTETOE604C - COMPUTER NETWORK

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 Which OSI layer is responsible for bit-by-bit data transmission?

- A) Data Link Layer
- B) Network Layer
- C) Transport Layer
- D) Physical Layer

Which transmission medium uses light to transmit data?

- A) Coaxial cable
- B) Microwave
- C) Optical fiber
- D) Twisted pair

3 The TCP/IP model has how many layers?

- A) 5
- B) 4
- C) 6
- D) 7

In packet switching, data is sent in:

- A) A continuous stream
- B) Circuits
- C) Fixed-sized cells
- D) Small packets independently

Which protocol is used in point-to-point data link connections?

- A) Ethernet
- B) PPP
- C) IP
- D) ARP

6 The main function of the Data Link Layer is:

- A) Addressing
- B) Routing
- C) Error detection and flow control
- D) Encryption

7 **Which access method is used in traditional Ethernet?**

- A) Token passing
- B) Polling
- C) CSMA/CD
- D) CSMA/CA

8 **Which Ethernet standard operates at 100 Mbps?**

- A) Standard Ethernet
- B) Gigabit Ethernet
- C) 10 Gigabit Ethernet
- D) Fast Ethernet

Which of the following uses a 128-bit address format?

- A) IPv4
- B) IPv6
- C) ARP
- D) ICMP

10 **Which protocol offers connectionless communication?**

- A) TCP
- B) UDP
- C) FTP
- D) SCTP

11 **TCP provides:**

- A) Unreliable, unordered delivery
- B) Reliable, ordered delivery
- C) Broadcasting service
- D) None of the above

Which application-layer protocol is used for email delivery?

- A) FTP
- B) HTTP
- C) SMTP
- D) SNMP

Q2. Solve the following.

- A) Compare the OSI model with the TCP/IP protocol suite, highlighting key differences. 6
- B) Explain various types of guided transmission media with their advantages and disadvantages. 6

Q3. Solve the following.

- A) Differentiate between LAN, MAN, and WAN with suitable examples. 6
- B) Compare various Random Access protocols based on their method, collision handling, and efficiency? 6

Q4. Solve Any Two of the following.

- A) Explain the structure and types of frames used in HDLC? 6
- B) Compare IPv4 and IPv6 in terms of addressing, header format, and features? 6
- C) What is Bluetooth? Explain its Architecture? 6

Q5. Solve Any Two of the following.

- A) Differentiate between TCP, UDP, and SCTP in terms of connection type, reliability, and application areas. 6
- B) Write short notes on 6
 - 1. FTP
 - 2. HTTP

- C) What is congestion control in networks? Explain Leaky Bucket and Token Bucket algorithms with diagrams. 6

Q6. Solve Any Two of the following.

- A) What is a firewall? Discuss different types of firewalls and their roles in network security? 6

- B) What is SNMP? Explain its architecture and how it is used for network management? 6

Explain the OSI reference model in detail. Describe the function of each layer with a suitable example? 6

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